



Master Specification: Vendors & Individuals (Service Requesters)

Role Tier: External Input

Objective: To request transportation of goods and specify required manual labor/resources for the move.

1. The Booking Interface (The Request Hub)

This is where the user defines the "System Input."

- **Item Categorization:**
 - **Commercial (Vendor):** High-volume, palletized goods, recurring shipments.
 - **Personal (Individual):** Luggage, furniture, household "shifts," or specialized goods.
- **Service Customization (The Add-ons):**
 - **Packing Service:** A toggle switch to request a "Packing Team." This triggers a notification to the **Warehouse Manager** to assign staff.
 - **Packing Materials (Utensils):** Selection of boxes, bubble wrap, or specialized crates.
 - **Labor Support:** Selection of the number of "Helper Labors" needed for loading/unloading.

2. Resource Impact on Other Roles

When an individual selects these add-ons, it changes the requirements for the rest of your system:

- **Impact on Dispatcher:** The system must now assign a vehicle with "Extra Seats" for the labor team and calculate a longer "Dwell Time" (since packing takes longer than just picking up a box).
- **Impact on Warehouse Manager:** They must ensure the inventory of "Packing Materials" is subtracted and the "Labor Staff" are scheduled for field duty instead of warehouse duty.
- **Impact on Driver:** The Driver becomes a "Team Lead" for the field staff assigned to that specific move.

3. Dynamic Pricing & Quotation Engine

Since the requirements vary, the system must calculate costs in real-time.

- **The Formula:**
$$\\$\\text{Total Cost} = \\text{Base Transport} + (\\text{Labor Count} \\times \\text{Hourly Rate}) + \\text{Material Costs}\\$$$
- **Instant Quote:** The user receives a price immediately based on the volume of goods and the services selected.

4. Generative AI Integration (The "Estimator")

This is where GenAI adds massive value to the individual user:

- **AI Visual Volume Estimator:** The user can upload a photo of their room/luggage. The AI analyzes the image and says: *"Based on the photo, you will need 5 Large Boxes and 2 Labors. Would you like me to add these to your booking?"*
- **Smart Scheduling:** If a user asks for a packing team, the AI checks the "Staff Availability" and suggests a time when the most experienced packers are free.
- **Inventory Chatbot:** The individual can ask, *"Do you have special crates for fragile glass utensils?"* and the AI confirms availability and adds it to the order.

Updated System Flow (The "Move" Lifecycle)

1. **Individual** books a "House Shift" + **Packing Team** + **Utensils**.
2. **Logistics Manager** sees the high-value transaction.
3. **Warehouse Manager** prepares the requested packing materials.
4. **Dispatcher** assigns a large van and 2 labors to the **Driver**.
5. **Driver & Labor Team** arrive, pack the goods, and take "Before/After" photos for **Customer Support** records.
6. **Customer Support Bot** keeps the individual updated on the team's arrival time.

Note if we add labours 🎉

To help your team integrate the **Individual/Moving Service** (with Labor and Packing) into your project, here is the breakdown of exactly what features you need to "Add-on" to each role's dashboard.

1. Logistics Manager (Owner)

Focus: Revenue from services and staff overhead.

- **Service Rate Manager:** A settings panel to set the price for **Labor per hour** and the rental/sale price of **Utensils** (crates, boxes, padding).
- **Labor Payroll Dashboard:** A view to see how many "Field Hours" the warehouse laborers have worked to calculate their wages.
- **Resource Analytics:** A chart comparing "Standard Delivery" revenue vs. "Personal Move/Shift" revenue to see which is more profitable.

2. Warehouse Manager (The Resource Provider)

Focus: Managing the "Physical Team" and the "Moving Supplies."

- **Staff Roster:** A live status list of all laborers (Status: *In Warehouse, On-Field with Driver, Off-Duty*).

- **Service Assignment:** When a "Moving" order comes in, the WM clicks a button to assign specific **Laborers** to that order.
- **Utensil Inventory:** A dedicated stock list for moving supplies (Bubble wrap, Crates, Tapes) that are consumed when an individual books a packing service.

3. Dispatcher (The Resource Optimizer)

Focus: Logistics of "People + Cargo."

- **Vehicle Seat Matching:** When assigning a truck, the Dispatcher must see the **Seat Capacity**. (e.g., You cannot assign 4 laborers if the truck only has 2 seats).
- **Service Time Blocking:** The system must automatically "block" a larger time window on the map for moves. (e.g., A "Shift" order blocks 3 hours, whereas a "Parcel" blocks 10 minutes).
- **Crew Manifest:** The ability to see which Driver is carrying which Laborers.

4. Driver (The Team Lead)

Focus: Field management of the assigned crew.

- **Crew Check-In:** A screen showing the names of the assigned laborers. The driver "clocks them in" when they board the truck.
- **Service Checklist:** A list of the requested services (e.g., "Pack Kitchen Utensils," "Move Sofa to 2nd Floor") that must be checked off before the customer signs.
- **Labor Attendance:** A button to report if a laborer did not show up or was sent back early.

5. Customer Support (The AI Bot)

Focus: Quoting and modifying service requests.

- **Instant Quote Engine:** The bot calculates the price based on: *Distance + No. of Laborers + Volume of Utensils*.
- **Real-Time Crew Tracking:** The bot answers: "*Where is my packing team?*" by checking the Driver's GPS (since the team is in the truck).
- **Service Add-on Modification:** Allows the customer to chat: "*Add 1 more helper for my move tomorrow,*" and the bot updates the bill and alerts the Warehouse Manager.

6. Vendor / Individual (The Client)

Focus: Specifying the "Shift" requirements.

- **"The Move" Booking Form:** A specialized UI where the user selects:
 - **Cargo Type:** (e.g., Household Goods, Luggage, Fragile Utensils).
 - **Labor Selection:** A counter to choose how many helpers are needed.
 - **Packing Toggle:** A switch for "We pack for you" or "I will pack myself."

 **Technical Tip: How to link them**

In your database, you need to link the Laborers to the Order.

Example: Order #101 is linked to Driver A **AND** Laborer 1, Laborer 2.